LAB5 Individual Exercises

Perspective projection and vanishing points | VNAV2021

1. Projection of a sphere

Take the image plane as z = 1, the optical center as the origin, and normalized image coordinates (u, v). A ray through (u, v) has direction w = (u, v, 1). For a sphere centered at C = (0, 0, d), with radius r and d > r + 1, a boundary ray is tangent when its distance from C equals r.

$$\text{distance}^2 = d^2(u^2 + v^2) / (1 + u^2 + v^2) = r^2$$

Solving for the image boundary gives

$$u^2 + v^2 = r^2 / (d^2 - r^2).$$

The projection is a filled circle centered at the principal point. Its radius is $r / \text{sqrt}(d^2 - r^2)$. This is slightly larger than the simple r/d approximation because the silhouette rays are tangent to the sphere.

Off-axis center

Now take C = (a, b, d), still with d > r so the sphere stays in front of the camera. The tangent-ray condition becomes

$$(au + bv + d)^2 = (a^2 + b^2 + d^2 - r^2)(u^2 + v^2 + 1).$$

Let $\rho = \text{sqrt}(a^2 + b^2)$, $A = d^2 - r^2$, $D = \rho^2 + A$, $x = (au + bv)/\rho$, and $y = (-bu + av)/\rho$. Completing the square yields

$$(x - \rho d/A)^2 / (r^2 D/A^2) + y^2 / (r^2/A) = 1.$$

Thus the off-axis silhouette is an ellipse. Its semiaxes are $r \text{ sqrt}(D)/A$ along the displacement direction and $r/\text{sqrt}(A)$ perpendicular to it. When a = b = 0, they become equal and the ellipse reduces to the on-axis circle.

2. Vanishing points

Let two distinct 3D lines share direction $q = (q_x, q_y, q_z)$. Points on line i are $P_i(\lambda) = P_{i0} + \lambda q$. Perspective projection gives

$$u_i(\lambda) = (X_{i0} + \lambda q_x) / (Z_{i0} + \lambda q_z),$$
$$v_i(\lambda) = (Y_{i0} + \lambda q_y) / (Z_{i0} + \lambda q_z).$$

If $q_z \neq 0$, taking λ to infinity gives the same finite point for both lines, regardless of their starting points:

$$(u_\infty, v_\infty) = (q_x/q_z, q_y/q_z).$$

This is the vanishing point. In homogeneous image coordinates it is $[q_x : q_y : q_z]$.

When do the projections remain parallel?

If $q_z = 0$, the depth of each point on a given line is constant. Its image trajectory is

$$(u_i(\lambda), v_i(\lambda)) = (X_{i0}/Z_{i0}, Y_{i0}/Z_{i0}) + \lambda(q_x/Z_{i0}, q_y/Z_{i0}).$$

Both image directions are proportional to (q_x, q_y), so the projected lines remain parallel (or coincide in a degenerate view). Their vanishing point is at infinity, $[q_x : q_y : 0]$. Therefore the required condition is $q_z = 0$: the 3D direction is parallel to the image plane.

Problem source: MIT 16.485 VNAV, LAB5 Exercises (2023), vnav.mit.edu/labs_2023/lab5/exercises.html.